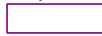
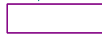


Daisy



File: Seed3-DevKit-Desktop_1.kicad_sch

Peripherals



File: Seed3-DevKit-Desktop_2.kicad_sch

Audio



File: audio.kicad_sch

POWER

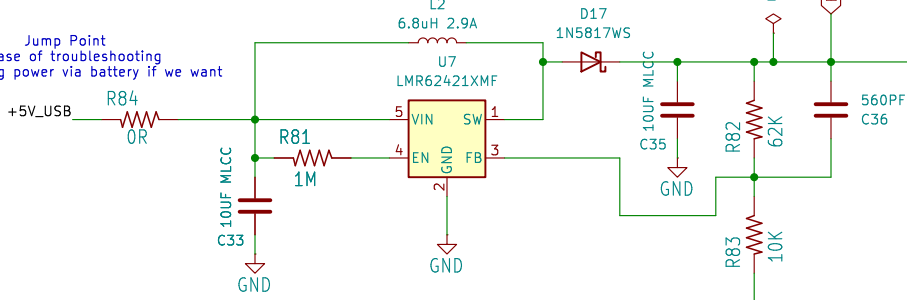
Selecting R2 with formula:
 $R2 = (V_{out}/V_{ref} - 1) \times R1$
 Where $V_{ref} = 1.255V$

For V_{out} : where $R_1 = 10K$ and $R_2 = 62K$
 $(R_2/R_1)+1 * V_{ref} = V_{out}$

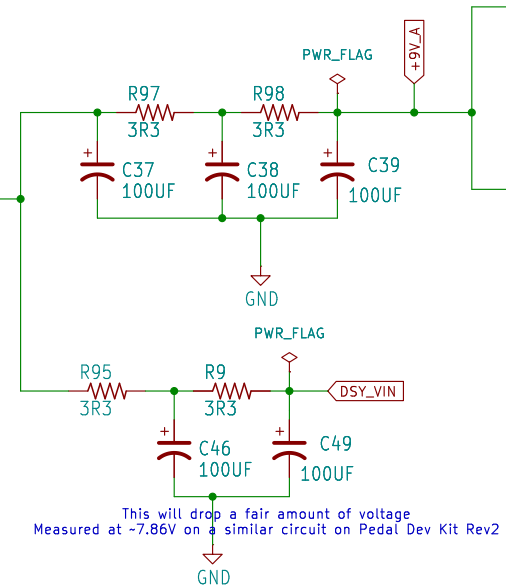
$$V_{out} = 9.036V$$

(or if my algebra was wrong, it would be.. 8.781V)
Either is totally fine .. and they're pretty close in this case.

Jump Point
In case of troubleshooting
or for testing power via battery if we want



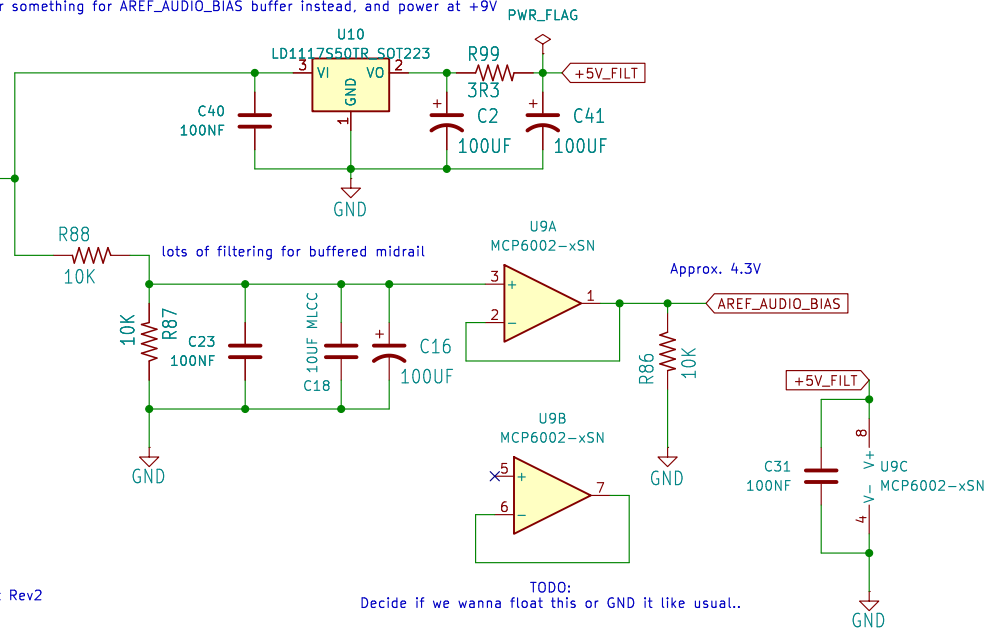
This will drop a fair amount of voltage
Measured at ~8.6V on similar circuit on Pedal Dev Kit Rev2



- At the moment, the +5V is powering
 - Headphone Amp
 - DAC output op-amp
 - Reference voltage for CV inputs
 - 6002 buffer for AREF_AUDIO_BIAS

We can remove this, and simplify slightly if we:

- Change DAC output op-amp to a rail-to-rail opamp that supports higher voltage (i.e. OPA1652)
- Power DAC output op-amp from +9V_A
- Add LM4040 for +5V reference for CV inputs
- Use TL074 or something for AREF_AUDIO_BIAS buffer instead, and power at +9V



Decide if we wanna float this or GND it like usual..

open hardware

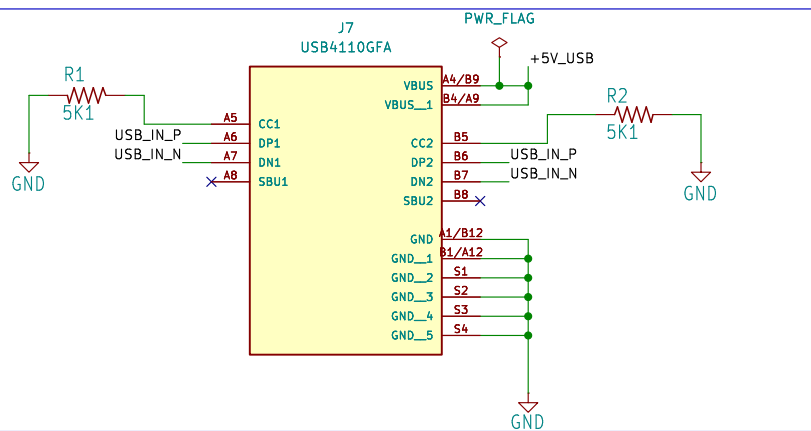
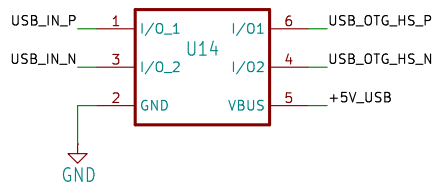
QR1



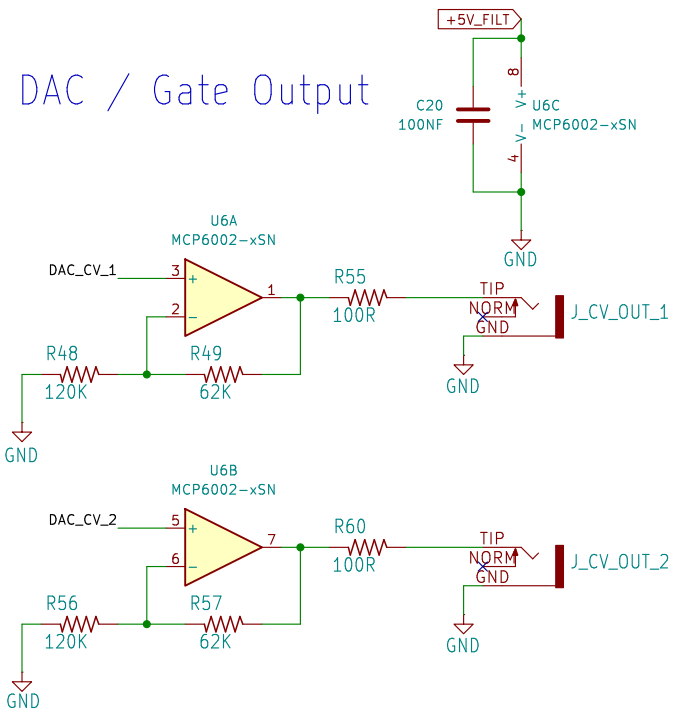
○ ○ ○

- H1 MOUNT-PAD-ROUND3.5
- H2 MOUNT-PAD-ROUND3.5
- H3 MOUNT-PAD-ROUND3.5
- H4 MOUNT-PAD-ROUND3.5

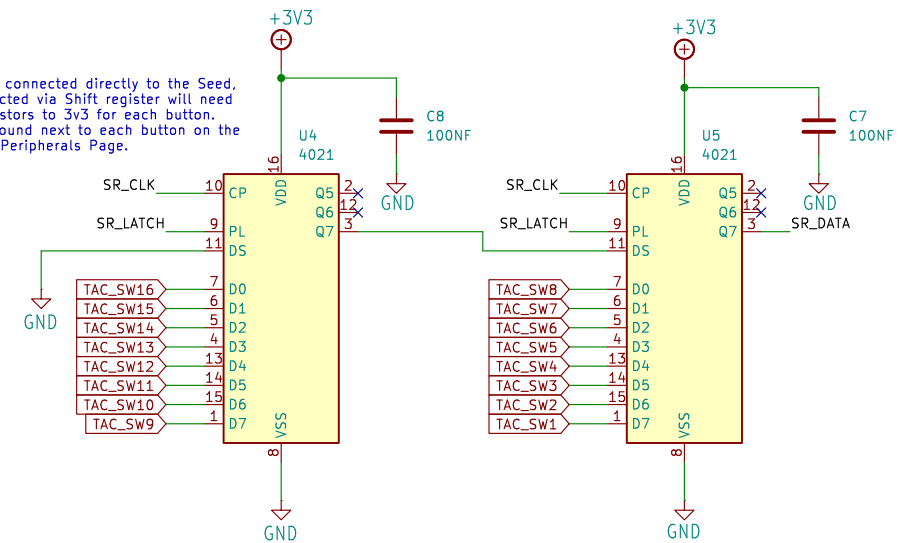
USB C



DAC / Gate Output

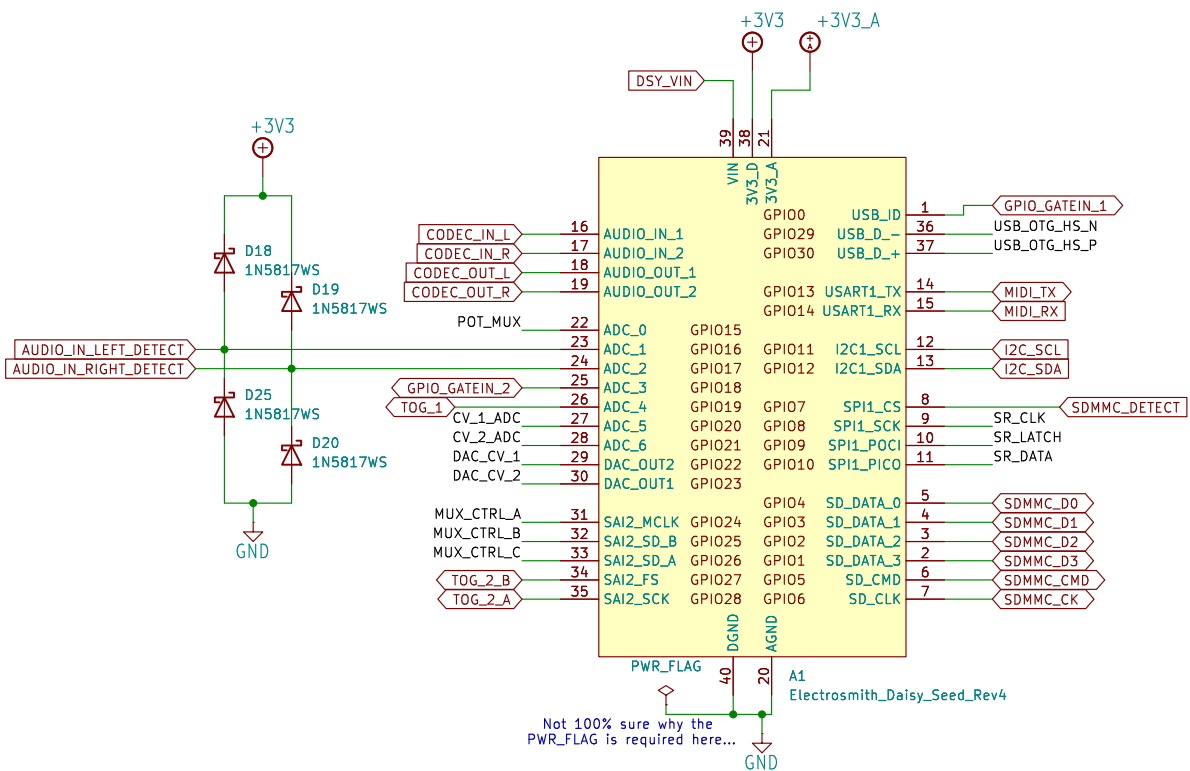
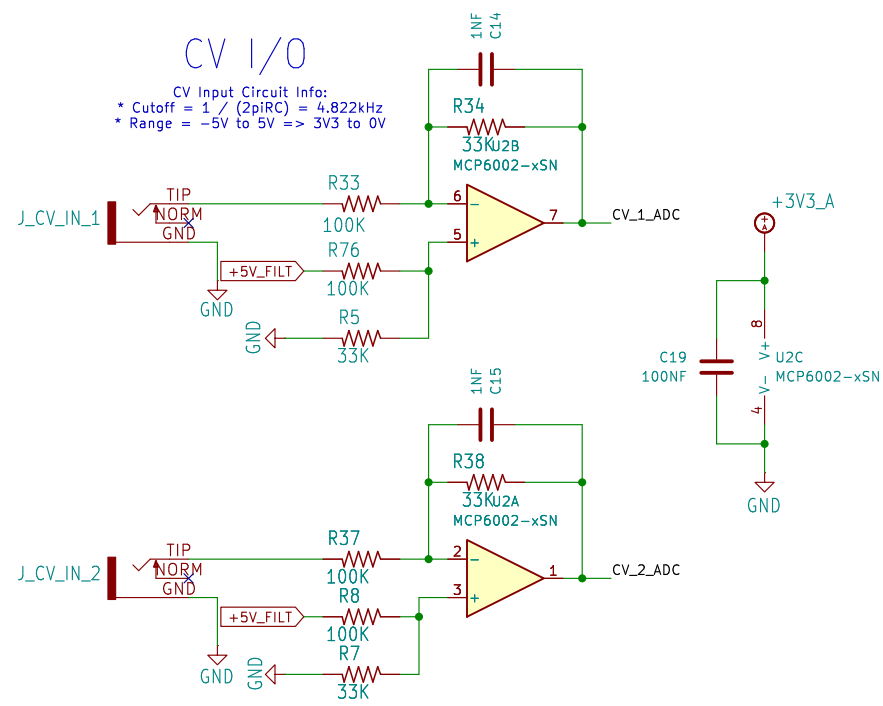


Unlike buttons connected directly to the Seed, buttons connected via Shift register will need pull up resistors to 3v3 for each button. These can be found next to each button on the Peripherals Page.



CV 1/0

CV Input Circuit Info:
 * Cutoff = $1 / (2\pi RC) = 4.8$
 * Range = $-5V$ to $5V \Rightarrow 3V3$



Not 100% sure why the PWR_FLAG is required here...

The schematic diagram illustrates the TAC SW module. At the top, a horizontal rail is connected to a +3V3_A supply. This rail contains eight potentiometers (POI 1 through POI 8) and eight variable resistors (VR 1 through VR 8). Each potentiometer is connected to the +3V3_A supply, and each variable resistor is connected to ground. Below this rail, there are 16 individual switch circuits, each consisting of a PTH Mini Button (S1 through S16) connected to a +3V3 supply through a 10K resistor (R1 through R16) and to ground through a 10K resistor (R17 through R32). The switches are labeled TAC_SW1 through TAC_SW16.

LEDs

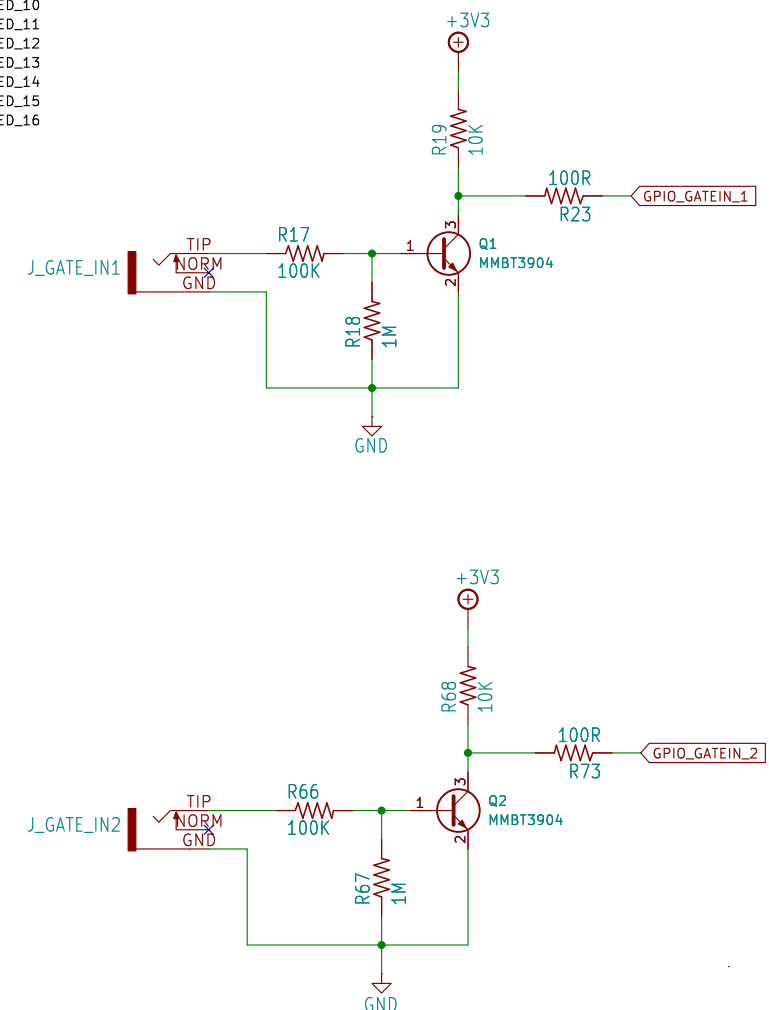
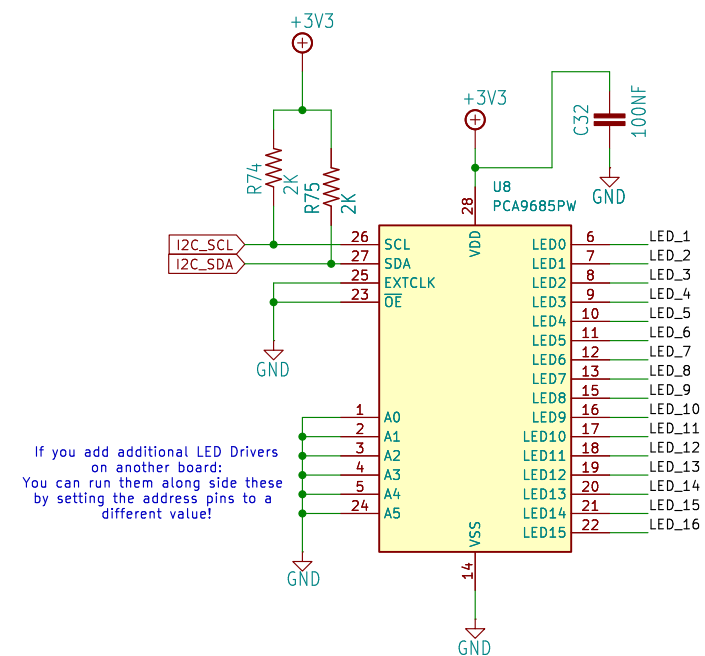
The diagram illustrates 16 individual LED connections, labeled LED_1 through LED_16. Each connection consists of a 2K resistor (R43, R40, R45, R69, R35, R41, R46, R70, R36, R42, R47, R71, R39, R44, R58, R72) connected in series with an LED (D1, D5, D9, D13, D2, D6, D10, D14, D3, D7, D11, D15, D4, D8, D12, D16). The LEDs are connected to ground (GND).

MIDI

The diagrams show three different MIDI connection configurations:

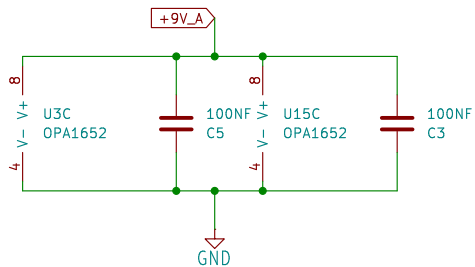
- Top Diagram:** MIDI IN (MIDI_IN_4 and MIDI_IN_5) connected to MIDI RX via a 74VHC11L1 inverter (U11). The inverter is powered by +3V3 and has a 100nF capacitor (C1) to ground. A 220R resistor (R4) is on the input line.
- Bottom Left Diagram:** MIDI IN connected to MIDI IN_4 and MIDI IN_5 via a 74VHC11L1 inverter (J10). The inverter is powered by +3V3 and has a 100nF capacitor (C34) to ground. A 100nF capacitor (C47) is on the output line. A PWR_FLAG signal is connected to the inverter's output.
- Bottom Right Diagram:** MIDI RX connected to MIDI THRU via a 74VHC11L1 inverter (J8). The inverter is powered by +3V3 and has a 100nF capacitor (C10) to ground. A 10R resistor (R29) is on the input line.

The PWR_FLAG signal is connected to the output of the inverter in the bottom left diagram. The text below the diagram states: "The PWR_FLAG feels awkward to have to do here.. But it is recommended to have the input isolated."

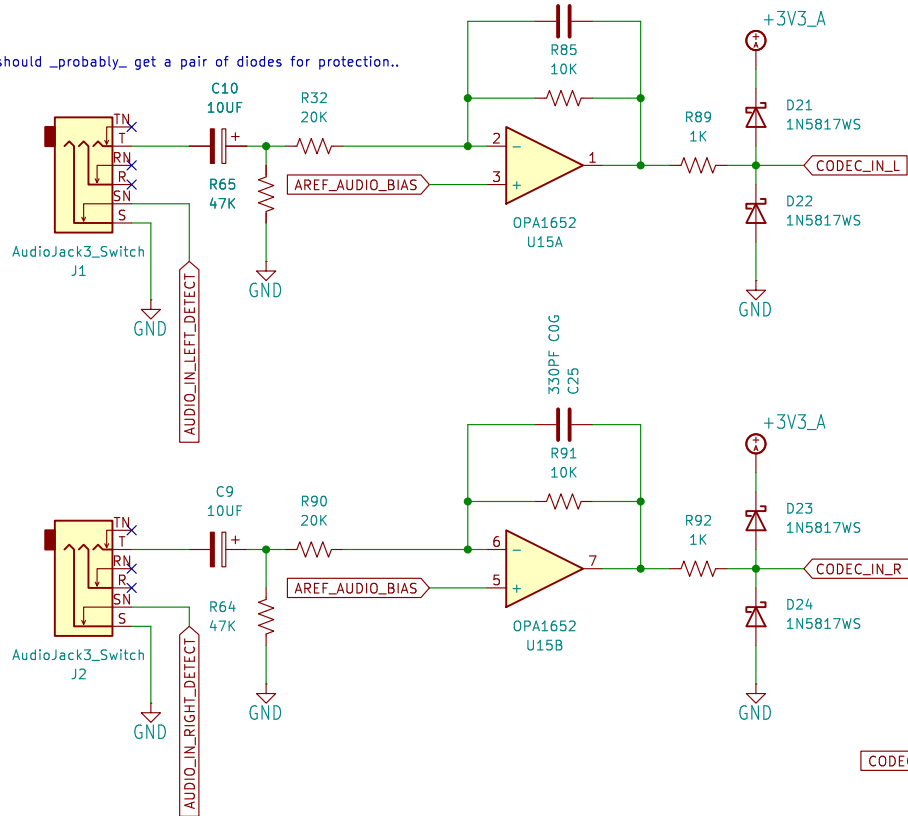


The diagram shows the connection between a microcontroller and an SD card (MICRO_SD7). The microcontroller pins are connected to the SD card pins through resistors. The SD card is labeled MICRO_SD7. The microcontroller pins are: SDMMC_CK (pin 5), SDMMC_CMD (pin 3), SDMMC_D0 (pin 7), SDMMC_D1 (pin 8), SDMMC_D2 (pin 1), SDMMC_D3 (pin 2). The SD card pins are: VDD (pin 4), CD (pin 1), MPB (pin 6), TF-01A (pin 7), GND (pin 8). The resistors are: R50 (47K), R51 (47K), R52 (47K), R53 (47K), R54 (47K). The power supply is +3V3.

AUDIO I/O



...s should _probably_ get a pair of diodes for protection..

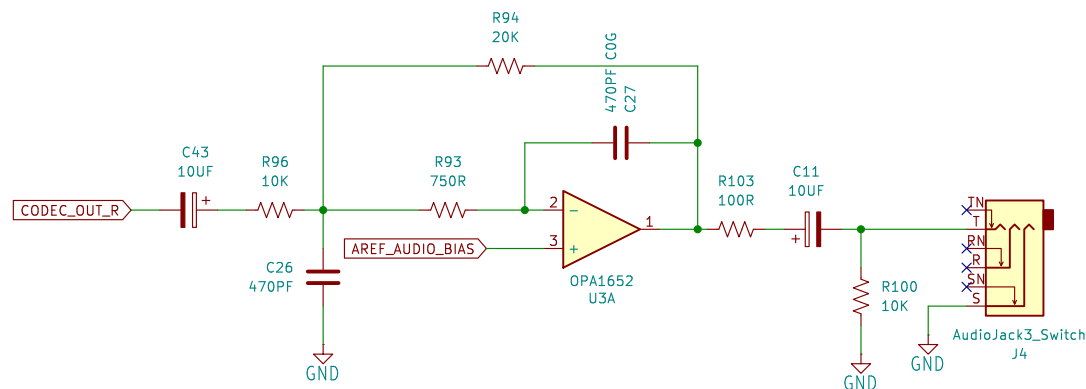


Gain is set to target +4dBu (maximum)
obviously this does not make it a pro-line-level device that way..
we'd need to probably have a certain amount of headroom.
If, instead we look at this as a -10dBV device, then we have
-14dB of headroom with these values.

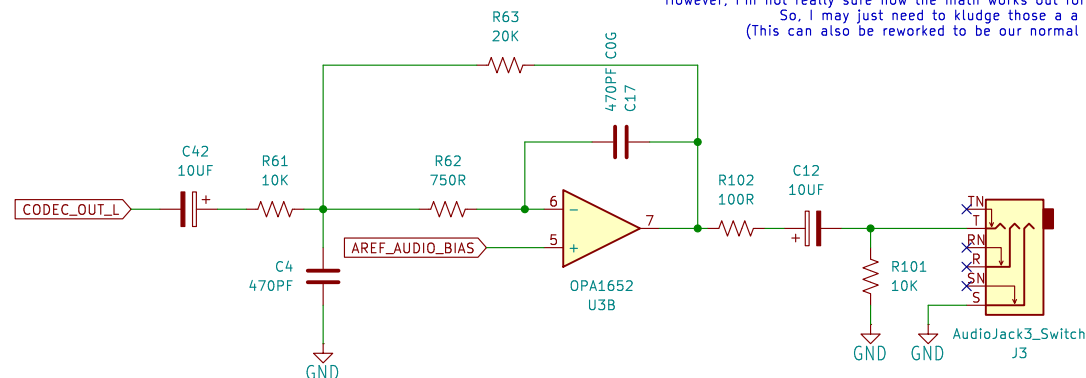
$G_{in}=0.814$ ($R_{in}=10k$, $R_{fb}=16k$)
 $G_{out}=1.228$ ($R_{in}=10k$, $R_{fb}=24k$)

Alternatively we can go closer to the PCM3060's "2Vrms" reference and just do:
 $G_{in}=0.5$, $G_{out}=2$:shrug:

For now, I've implemented the left channel of audio with less atten/gain.



I updated these to be gain of 2.
However, I'm not really sure how the math works out for the other two values..
So, I may just need to kludge those a bit IRL.
(This can also be reworked to be our normal basic filter)



DECISION - Two Choices:
1. Add LD1117 (or LP2985) or other 5V Linear Reg.
2. Power TPA6110 from Seed's 3v3A

